

Case Study: Hospital Management System

Insert Many Documents (MongoDB)

```
insertedIds: {  
  '0': ObjectId('6a12f7f79b28ef72a8a4257e'),  
  '1': ObjectId('6a12f7f79b28ef72a8a4257f'),  
  '2': ObjectId('6a12f7f79b28ef72a8a42580'),  
  '3': ObjectId('6a12f7f79b28ef72a8a42581'),  
  '4': ObjectId('6a12f7f79b28ef72a8a42582'),  
  '5': ObjectId('6a12f7f79b28ef72a8a42583'),  
  '6': ObjectId('6a12f7f79b28ef72a8a42584'),  
  '7': ObjectId('6a12f7f79b28ef72a8a42585'),  
  '8': ObjectId('6a12f7f79b28ef72a8a42586'),  
  '9': ObjectId('6a12f7f79b28ef72a8a42587')  
}
```

1. Find all patients whose treatment cost is greater than 20,000 and display only their name and department.

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```
> db.patients.aggregate([
  { $match: { treatmentCost: { $gt: 20000 } } },
  { $project: { _id: 0, patientName: 1, department: 1 } }
])
< {
  patientName: 'Meena',
  department: 'Neurology'
}
{
  patientName: 'Kiran',
  department: 'Neurology'
}
{
  patientName: 'Vijay',
  department: 'Cardiology'
}
{
  patientName: 'Priya',
  department: 'Neurology'
}
```

2. List all patients sorted by treatment cost in descending order.

```
> db.patients.aggregate([
  { $sort: { treatmentCost: -1 } }
])
```

3. Display the first 5 patient records.

```
> db.patients.aggregate([
  { $limit: 5 }
])
```

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4. Skip first 3 patients and display the remaining records.

```
> db.patients.aggregate([
  { $skip: 3 }
])
```

5. Find patients from the Cardiology department and display only their names.

```
> db.patients.aggregate([
  { $match: { department: "Cardiology" } },
  { $project: { _id: 0, patientName: 1 } }
])
```

6. Find total treatment cost per department.

```
> db.patients.aggregate([
  { $group: { _id: "$department", totalCost: { $sum: "$treatmentCost" } } }
])
```

7. Find average treatment cost per doctor and display results in ascending order.

```
> db.patients.aggregate([
  { $group: { _id: "$doctor", avgCost: { $avg: "$treatmentCost" } } },
  { $sort: { avgCost: 1 } }
])
```

8. Find departments where total treatment cost is greater than 50,000.

```
> db.patients.aggregate([
  { $group: { _id: "$department", totalCost: { $sum: "$treatmentCost" } } },
  { $match: { totalCost: { $gt: 50000 } } }
])
```

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9. Find number of patients in each department and sort the result in descending order.

```
> db.patients.aggregate([
  { $group: { _id: "$department", patientCount: { $sum: 1 } } },
  { $sort: { patientCount: -1 } }
])
```

10. Find top 3 doctors with highest total treatment cost.

```
> db.patients.aggregate([
  { $group: { _id: "$doctor", totalCost: { $sum: "$treatmentCost" } } },
  { $sort: { totalCost: -1 } },
  { $limit: 3 }
])
```

11. Find patients whose age is greater than 40, group them by department, and count number of patients.

```
> db.patients.aggregate([
  { $match: { age: { $gt: 40 } } },
  { $group: { _id: "$department", patientCount: { $sum: 1 } } }
])
```

12. Find average treatment cost per department, display only department and average cost, and sort descending.

```
> db.patients.aggregate([
  { $group: { _id: "$department", avgCost: { $avg: "$treatmentCost" } } },
  { $project: { _id: 1, avgCost: 1 } },
  { $sort: { avgCost: -1 } }
])
```

13. Find departments having more than 2 patients admitted.

```
> db.patients.aggregate([
  { $group: { _id: "$department", patientCount: { $sum: 1 } } },
  { $match: { patientCount: { $gt: 2 } } }
])
```

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14. Display top 3 departments based on highest total treatment cost after skipping the first one.

```
> db.patients.aggregate([
  { $group: { _id: "$department", totalCost: { $sum: "$treatmentCost" } } },
  { $sort: { totalCost: -1 } },
  { $skip: 1 },
  { $limit: 3 }
])
```

15. Find doctors treating patients with cost greater than 10,000, calculate total cost, and show only top 2 doctors.

```
> db.patients.aggregate([
  { $match: { treatmentCost: { $gt: 10000 } } },
  { $group: { _id: "$doctor", totalCost: { $sum: "$treatmentCost" } } },
  { $sort: { totalCost: -1 } },
  { $limit: 2 }
])
```